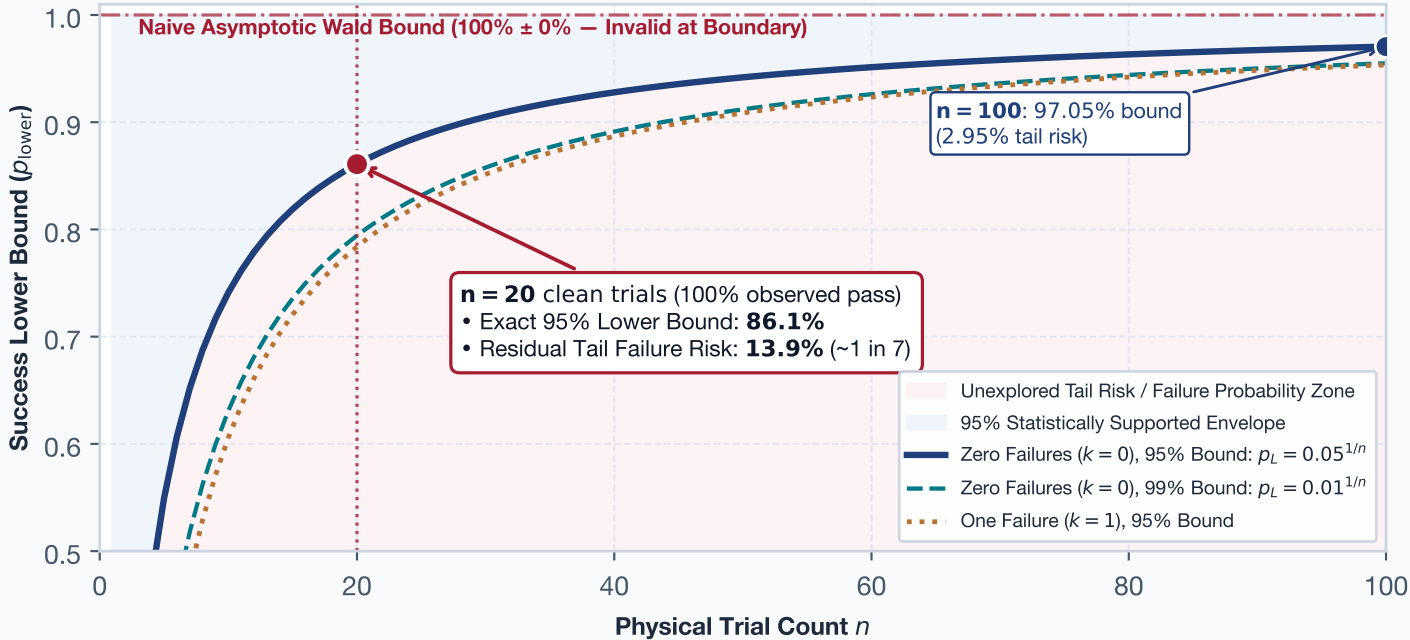
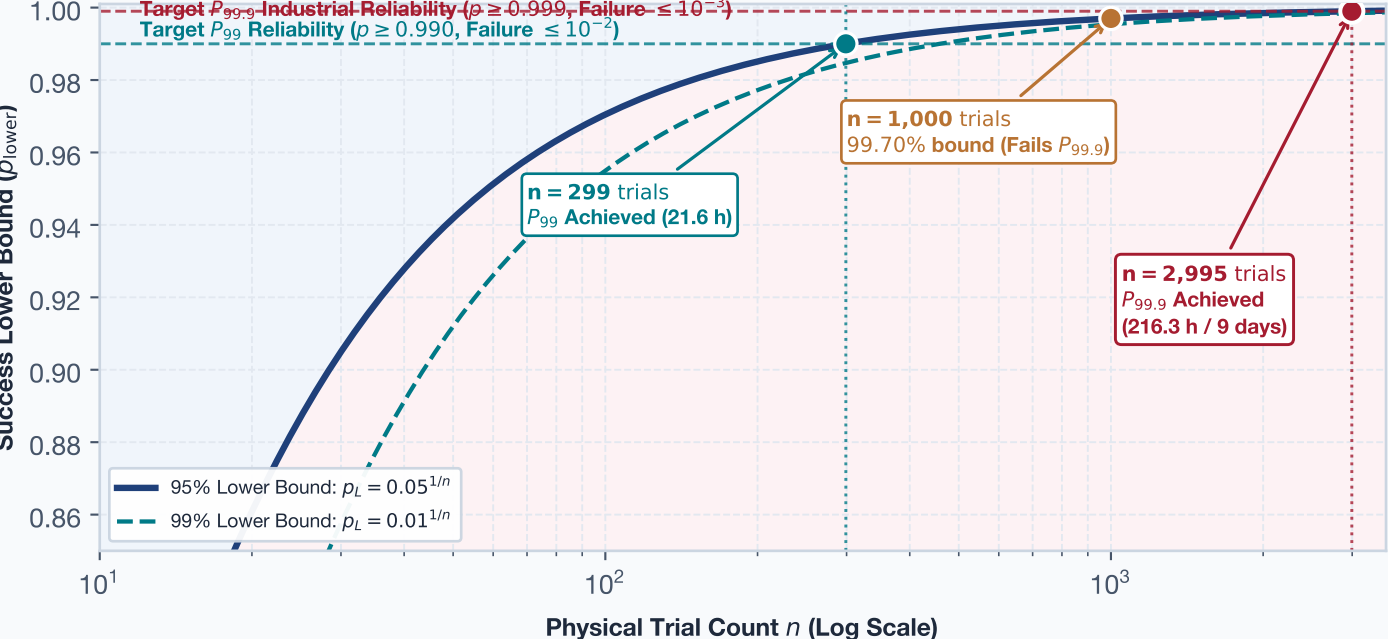


Non-Asymptotic Clopper-Pearson Bounds & Physical Testing Budget Limits

(a) Small-Sample Testing Regime & Tail Risk ($n \leq 100$)



(b) High-Reliability Asymptote & The Nonstationarity Barrier ($n \rightarrow 3,000$)



PHYSICAL TESTING BUDGET & STATISTICAL LIMITS

Trial unit: 50 s run + 120 s reset + 30 s zero + 60 s cooling = 260 s (4.33 min)

Trials (n)	95% Bound	Tail Risk	Wall-Clock Time	Hardware State
n = 20	86.10%	13.90% (1 in 7)	1.4 hours	Negligible wear
n = 100	97.05%	2.95% (1 in 34)	7.2 hours	Minor thermal rise
n = 299	99.00%	1.00% (1 in 100)	21.6 hours	Gear backlash shifts
n = 1,000	99.70%	0.30% (3 in 1000)	72.2 h (3.0 d)	Tire friction drops 15%
n = 2,995	99.90%	0.10% (1 in 1000)	216.3 h (9.0 d)	Exceeds bearing life!

THE THREE ARCHITECTURAL REMEDIES FOR PHYSICAL AI

1. Narrow Operational Domain (ODD)

Restrict machine to benign velocities and tight environmental bounds where modest sample counts ($n = 50$ -200) provide defensible statistical coverage.

2. Parallel HIL / Fleet Testing

Distribute runs across multiple test bays; beware that shared batch tolerances, floor finishes, and weather induce common-mode spatial correlations.

3. Enforce Deterministic Gate in Nervous System

Delegate safety invariants, CBFs, and emergency stops to trusted real-time MCU reflex loops (1 kHz), removing unachievable reliability claims from the learned Brain.